Ayanda Thabo Phaketsi

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Education

University of Cape Town Cape Town, South Africa
BSc Computer Science and Business Computing **Dec 2025**

Relevant Coursework: Software Engineering, Algorithms and Data Structures, Databases, Operating Systems, Computer Networks, Machine Learning, Systems Analysis and Design

Projects

Strat Edge Portal Pro — Project Management System **2026**
Developed for Strat Edge
Project Description: Built a portal for tracking project schedules and finances. The system includes portfolio monitoring and reporting for stakeholders.
Key Contributions:
- **System Build:** Created the platform using FastAPI and React with JWT authentication and access control.
- **Analytics:** Built backend services for project metrics, budget burndown, and task completion.
- **Storage:** Integrated document storage with Google Cloud Storage using signed URLs for access.
- **Dashboards:** Built charts and network diagrams to track project progress.

Technologies Used:
- **Backend:** Python (FastAPI), SQLAlchemy, Google Cloud Storage, Pandas
- **Frontend:** React (TypeScript), Vite, Tailwind CSS, Recharts

clia — Local AI Assistant for Software Tasks **2025**
`https://pypi.org/project/clia-swe-ai/`
Project Description: Built an AI assistant that helps with software tasks like refactoring and debugging by interacting with the local developer environment.
Key Contributions:
- **Logic:** Built a loop for managing tool usage and multi-turn chat with Google Gemini.
- **Context:** Used Model Context Protocol (MCP) to build a tool server for filesystem and command-line actions.
- **Safety:** Added a permission model where users approve destructive operations.
- **Tooling:** Created tools for code editing and codebase snapshots.

Technologies Used:
- **Languages/Frameworks:** Python, Google Gemini API, FastAPI, MCP
- **UI Tools:** Rich, prompt-toolkit, WebSockets

MemGallery — AI Memory Assistant for Android **2025**
`https://github.com/Past-da-king/MemGallery`
Project Description: Built an Android app that captures and processes media to enable semantic search, making AI features available on all Android 7.0+ devices.
Key Contributions:
- **Capture:** Built a system to detect and process screenshots automatically.
- **Processing:** Integrated Gemini 2.5 Flash to extract events and notes from images and audio.
- **Search:** Built a memory bank where users find items using natural language.
- **Interface:** Built the UI using Material Design 3 and Jetpack Compose.

Technologies Used:
- **Mobile Stack:** Kotlin, Jetpack Compose, Hilt, Room Database
- **System APIs:** WorkManager, ContentObservers, Google Generative AI SDK

Nexa — Offline Communication Platform **2025**
`https://github.com/Past-da-king/rednexa`
Project Description: Worked with a team to build a communication system for areas without internet. The app allows messaging through a peer-to-peer network.
Key Contributions:
- **Networking:** Built a custom protocol for data transfer via Bluetooth and Wi-Fi Direct.
- **Reliability:** Added logic for data deduplication and message delivery.
- **Security:** Integrated Google Tink for end-to-end encryption of messages.
- **Identity:** Created a local system for identity management without phone numbers.

Technologies Used:
- **Core Stack:** Kotlin, Google Nearby Connections, Google Tink, Jetpack Compose
- **Data Management:** Room, Kotlinx Serialization

StudyInc — Collaborative Productivity Tool **2025**
https://phaketsi.vercel.app/#projects
Project Description: Built a collaborative platform for study sessions. It includes real-time presence, timers, and analytics to help students track their work.
Key Contributions:

- **Syncing:** Used Socket.io for chat and timer sync, and WebRTC for video calls.
- **Desktop:** Used Electron to build features like overlays for timers and tasks.
- **Analytics:** Built a dashboard using React and Tailwind CSS to show study habits.
- **Backend:** Built a Node.js server with MongoDB and JWT authentication.

Technologies Used:

- **Tech Stack:** Node.js, Next.js, Electron, MongoDB, Socket.io, WebRTC
- **Styling:** Tailwind CSS, shadcn/ui

Verilookie — Deepfake and AI Detection **2025**
https://github.com/Maphuti-Shilabje/verilookie
Project Description: Built a web app to detect deepfakes and AI-generated media, combined with a learning platform.
Key Contributions:

- **Detection:** Integrated Hugging Face Vision Transformers and NVIDIA APIs for media analysis.
- **Education:** Built a quiz system powered by Gemini API that generates questions based on user history.
- **Optimization:** Used OpenCV for video frame sampling and Docker for containerization.

Technologies Used:

- **Backend:** Python (FastAPI), PyTorch, OpenCV, Docker
- **Frontend/AI:** React, Google Gemini API, Hugging Face Transformers

Awards & Achievements

- **Winner — Entelect University Challenge 2025:** Recognized for technical problem-solving.
- **2nd Place — Predictive Insights 2025 Hackathon:** Built a solution for data modeling.

Technical Skills

Languages: Java, Python, JavaScript, TypeScript, Kotlin, C#, SQL, LaTeX
Frameworks/Tools: React, Next.js, FastAPI, Node.js, Jetpack Compose, Electron, MongoDB, Docker, Git
Specializations: AI/LLM Integration, Distributed Systems, Full-Stack Mobile & Web